YNU
YOKOHAMA National University

Modeling of Si Rib and Photonic Crystal Mach-Zehnder Modulators for Electro-Optic Co-Simulation up to 64 Gbaud

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This work was supported through the activities of VDEC, The University of Tokyo,
in collaboration with Cadence Design Systems, Inc. and Keysight Technologies, Inc.

Background

Challenge

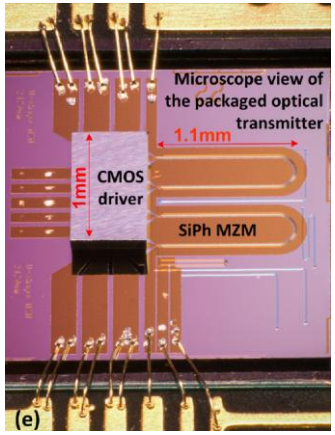
Huge data traffic for cloud services
High power consumption of interconnects

Beyond 400G

50–100-Gbaud modulation with a power of <300 mW
(Including Drivers)

Electro-Optic Hybrid Integrated Si Transmitters

Soton Univ. ’20



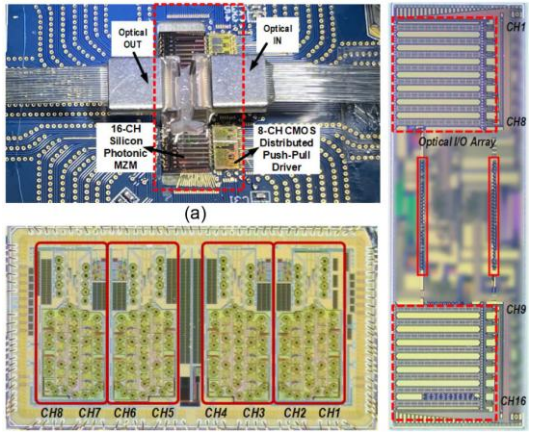
100-Gbps NRZ
203 mW/ch

STMicroelectronics ’20



53-Gbps PAM4 × 4ch
290 mW/ch

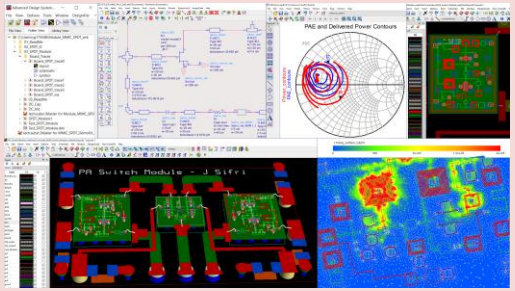
Peking Univ. ’23



50-Gbps NRZ × 16ch
268 mW/ch

Transceiver Design Flow

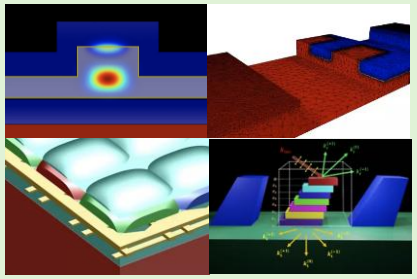
Electronics



© Keysight Technologies, Inc.

- HSPICE
- Spectre
- ADS
- etc…

Photonics



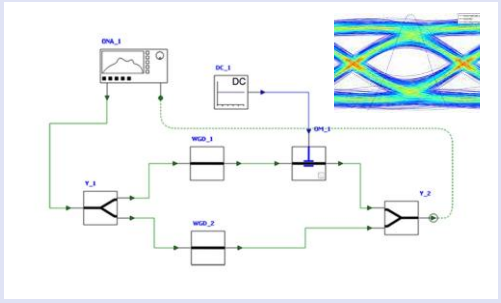
© Ansys, Inc.

- Lumerical
- Rsoft
- OptDesigner
- etc…

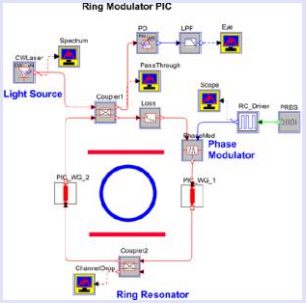
S-parameters

Models

Block Level Simulation



© Ansys, Inc.



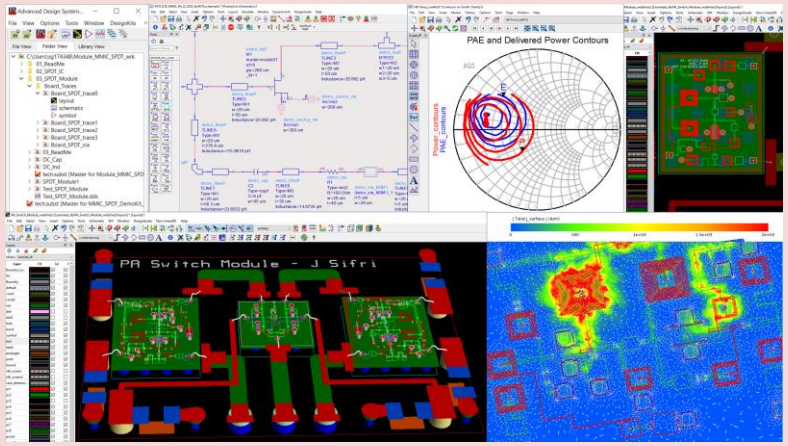
© Synopsys, Inc.

- Lumerical Interconnect
- OptSim Circuit
- OptiSpice
- MATLAB Simulink
- etc…

There is a barrier between electronics and photonics

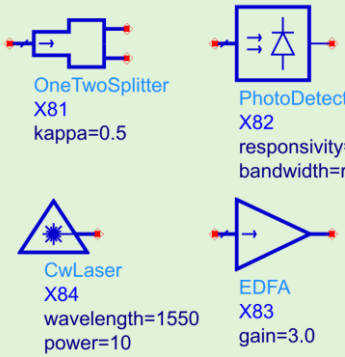
Verilog-A Modeling Approach

Electronics Simulators



© Keysight Technologies, Inc.

Photonics (Verilog-A PDKs)




```
`include "constants.vams"
`include "disciplines.vams"

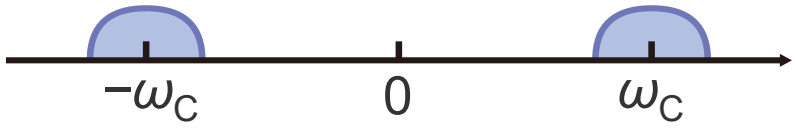
module Waveguide(in, out);
    input [0:1] in;
    output [0:1] out;
    optical [0:1] in, out,
```

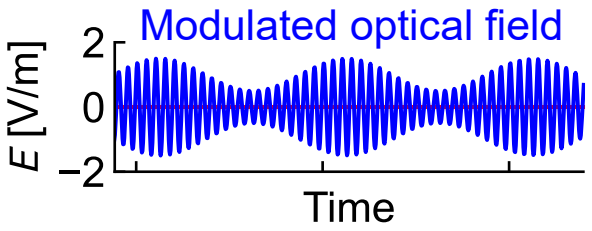
- Verilog-A
  - Analog circuit (or device) modeling language
  - Used in many electronic circuit simulators
- Photonics device models in electronics simulators
  - Allows for co-design and co-optimization

Verilog-A Model Overview

C. Sorace-Agaskar, *et al.*, *Opt. Express*, vol. 23, no. 21, pp. 27180–27203, Oct. 2015.

Real Signal

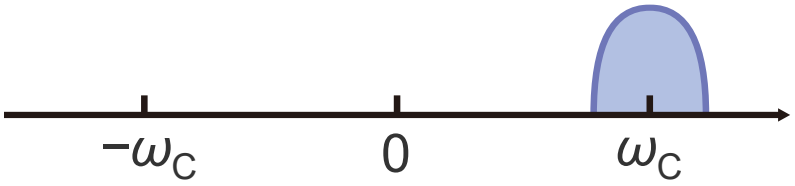


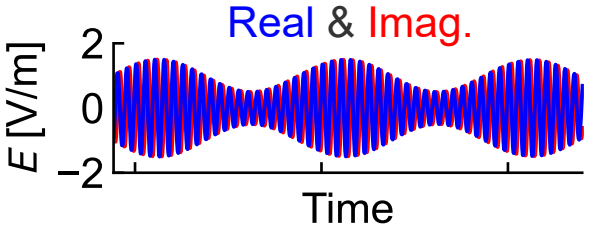


× Difficult to compute instantaneous amplitude and phase

× Optical frequency ω_C is too high (≈300 THz) for circuit simulation

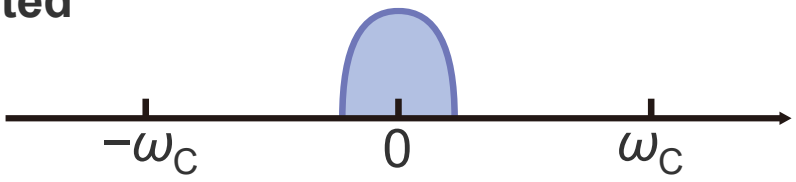
Analytic Signal

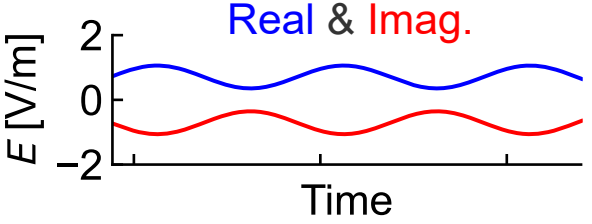




✓ Simple handling of amplitude and phase like $A(t)e^{j(\omega_C t+\theta(t))}$

Frequency Shifted Analytic Signal

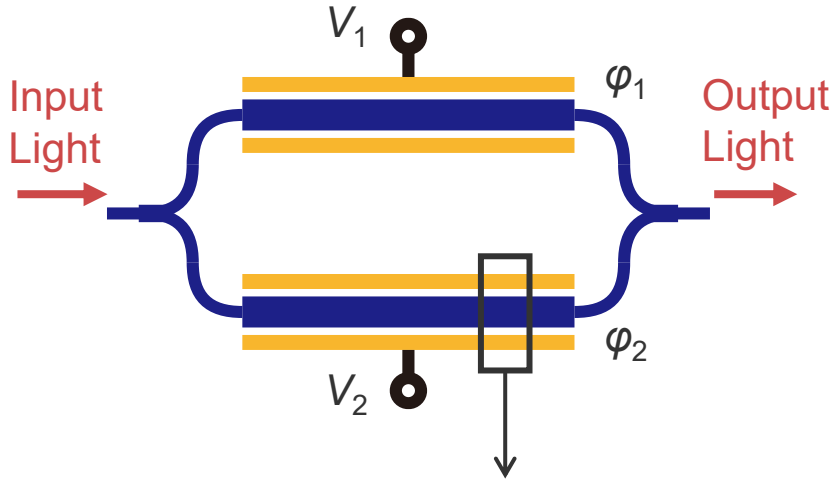


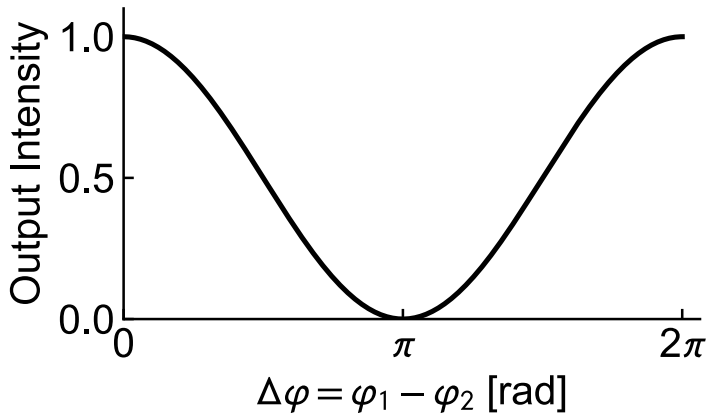


✓ Reduces simulation time steps

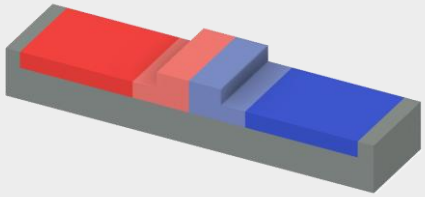
Si Mach-Zehnder Modulator

Modulators convert electrical signals to light intensity

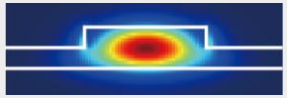




Rib Waveguide

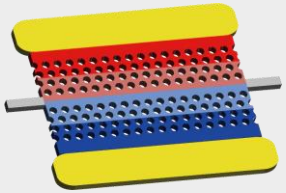


Cross Section

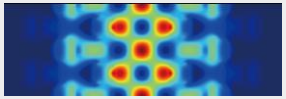


✓Commercial products

Photonic Crystal Waveguide (PCW)



Top View

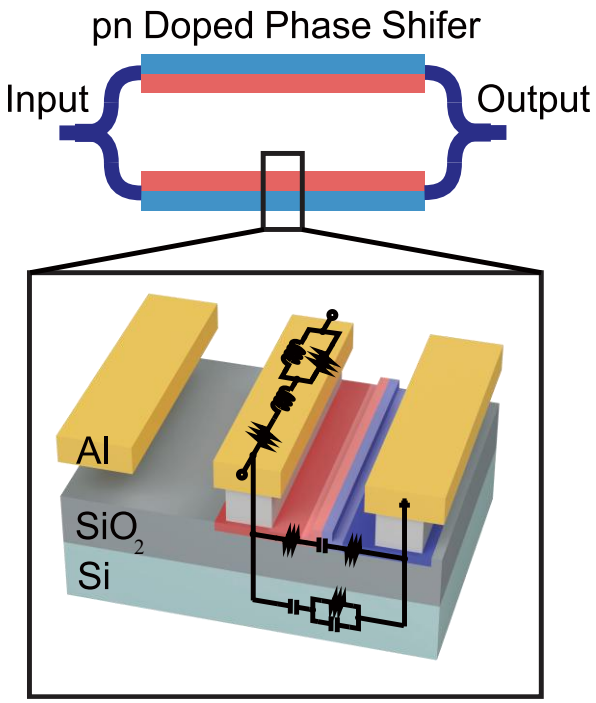


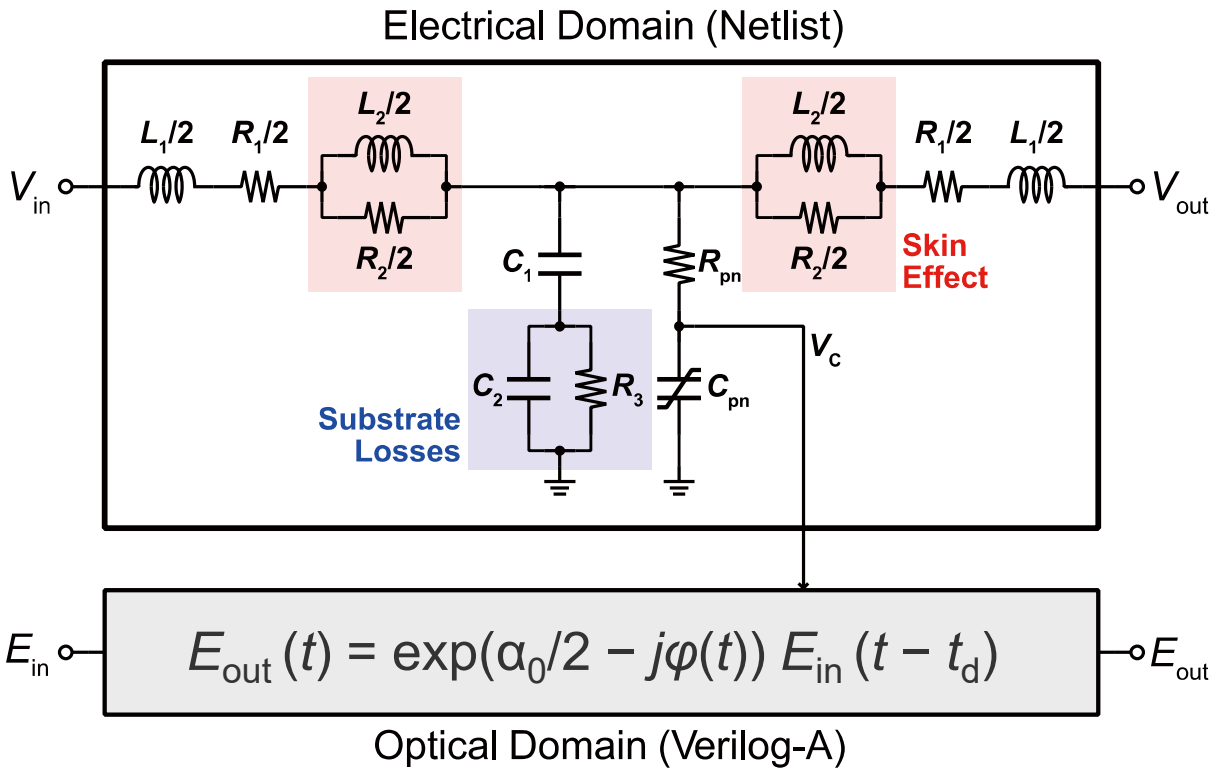
✓Efficient due to slow light effect

High-frequency (> 25 Gbaud) Verilog-A model is still missing

This work: Accurate RF modeling of Si rib and PCW modulators

Proposed Model Schematic

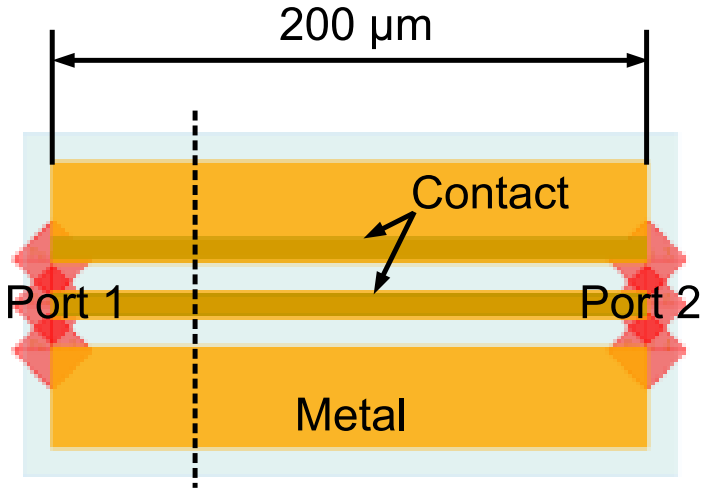


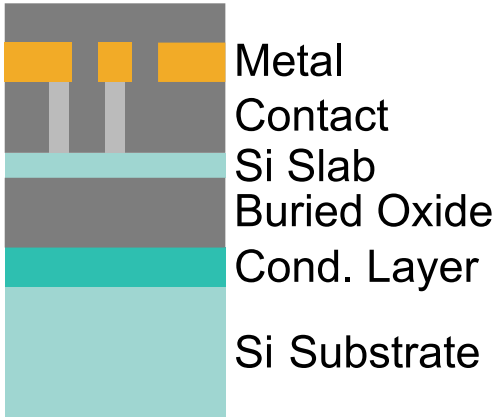


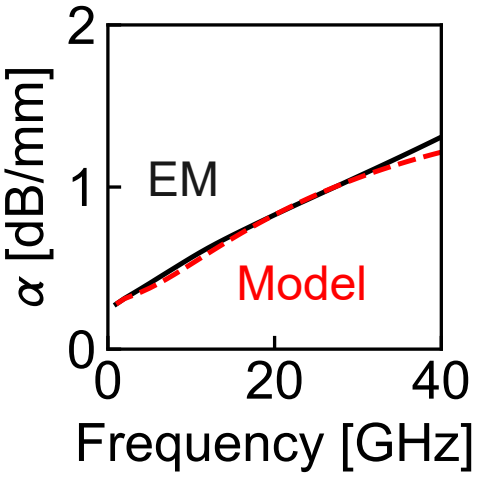
- Distributed circuit model
  - Provides the phase mismatch between RF and optical pulses
  - Provides the RF propagation losses (skin effect and substrate losses)
- Fitting parameters: L_1, L_2, R_1, R_2, C_1, C_2, R_3, R_{pn}, $C_{pn}(V)$, $n_{eq}(V)$, n_g

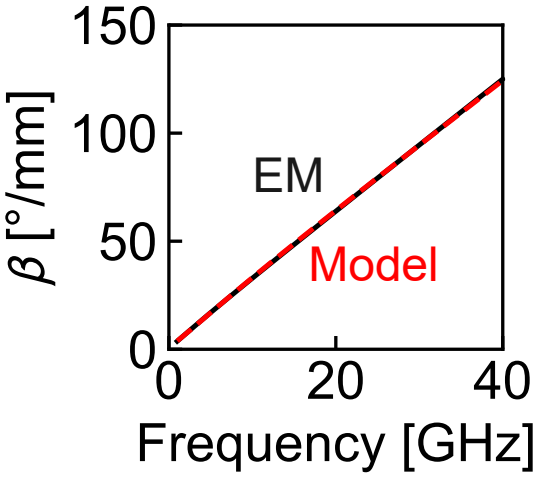
Electrode Extraction

- EM simulation with Keysight Momentum
- Electrodes only (not include pn junction)









2-port S-parameters

↓

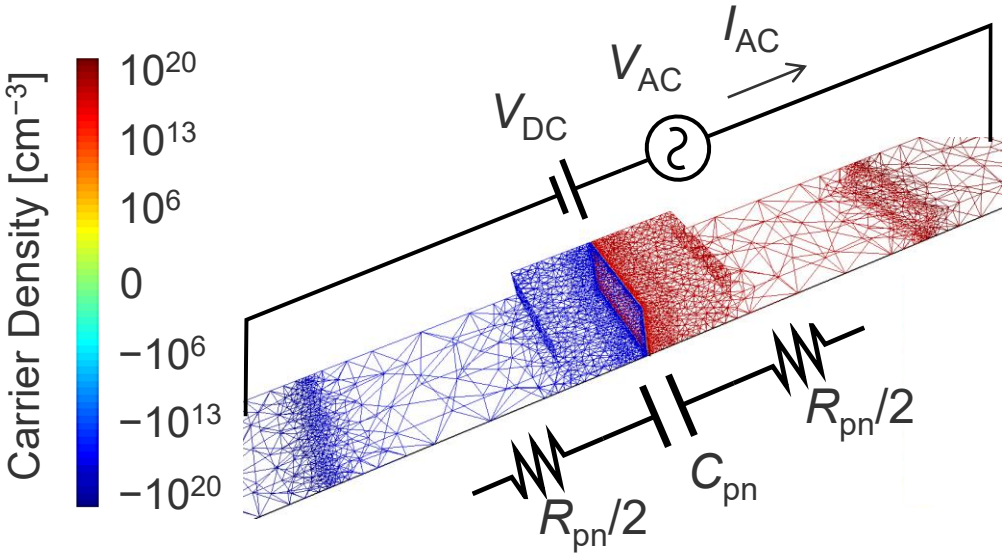
Impedance Z_0 and propagation constant $\gamma = \alpha + j\beta$

↓ Fitting

$L_1, L_2, R_1, R_2, C_1, C_2, R_3$

RC Time Constant

Small signal AC analysis using Lumerical CHARGE



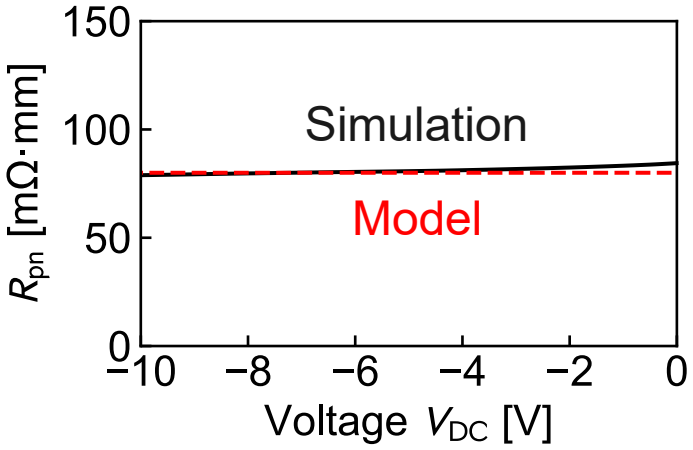
Calculate Z_{pn} for various V_{DC}

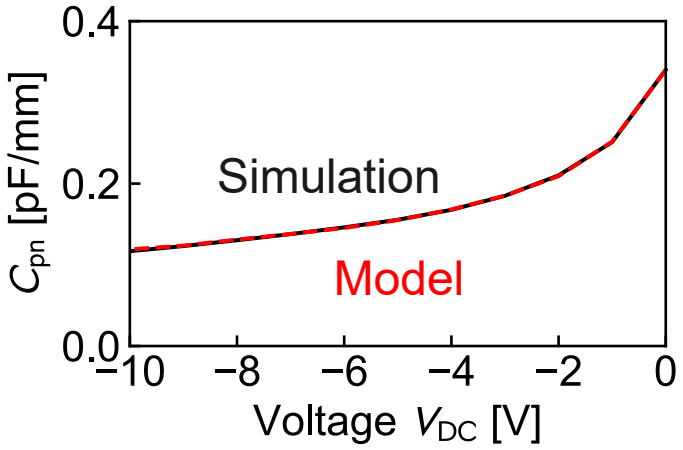
↓

R_{pn} = real (Z_{pn})
C_{pn} = 1 / ω | imag(Z_{pn}) |

↓ Fitting

$C_{pn}(V) = C_6V^6 + C_5V^5 + \ldots + C_0$

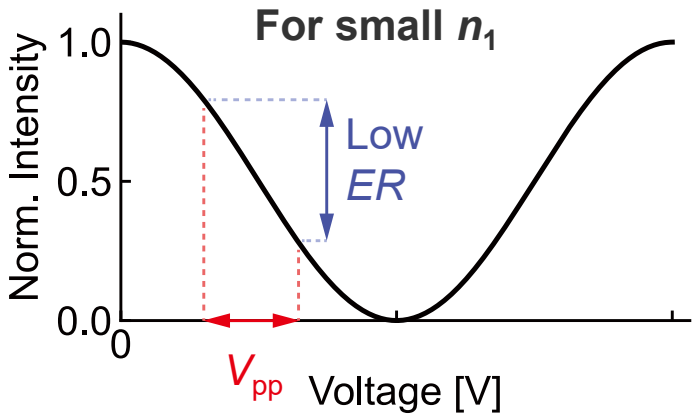


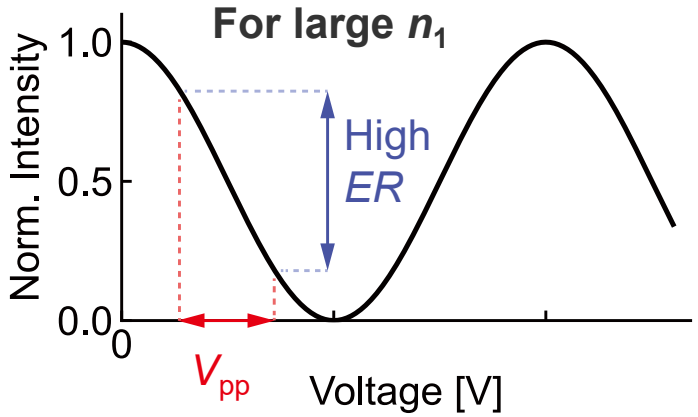


Modulation Efficiency

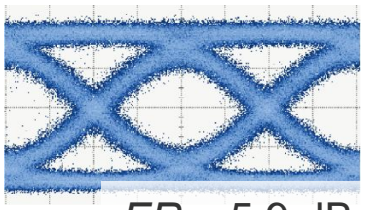
Modal equivalent index: $n_{eq}(V) = n_0 + n_1 V$

- n_0 was calculated using mode simulation (ex. Lumerical MODE)
- n_1 was determined from the *ER* of measured eye



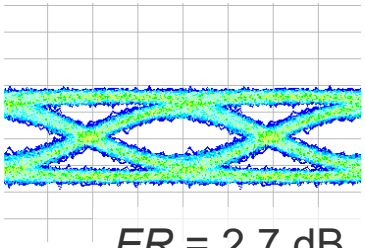


Measurement



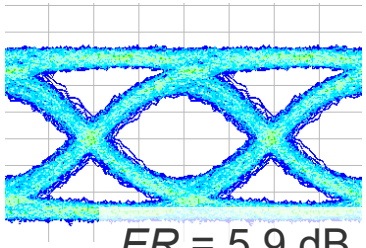
ER = 5.9 dB

Too small n_1



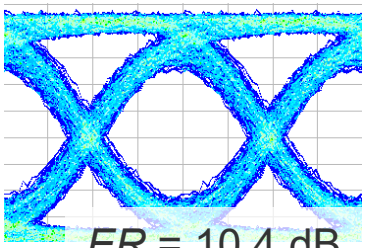
ER = 2.7 dB

Optimal n_1



ER = 5.9 dB

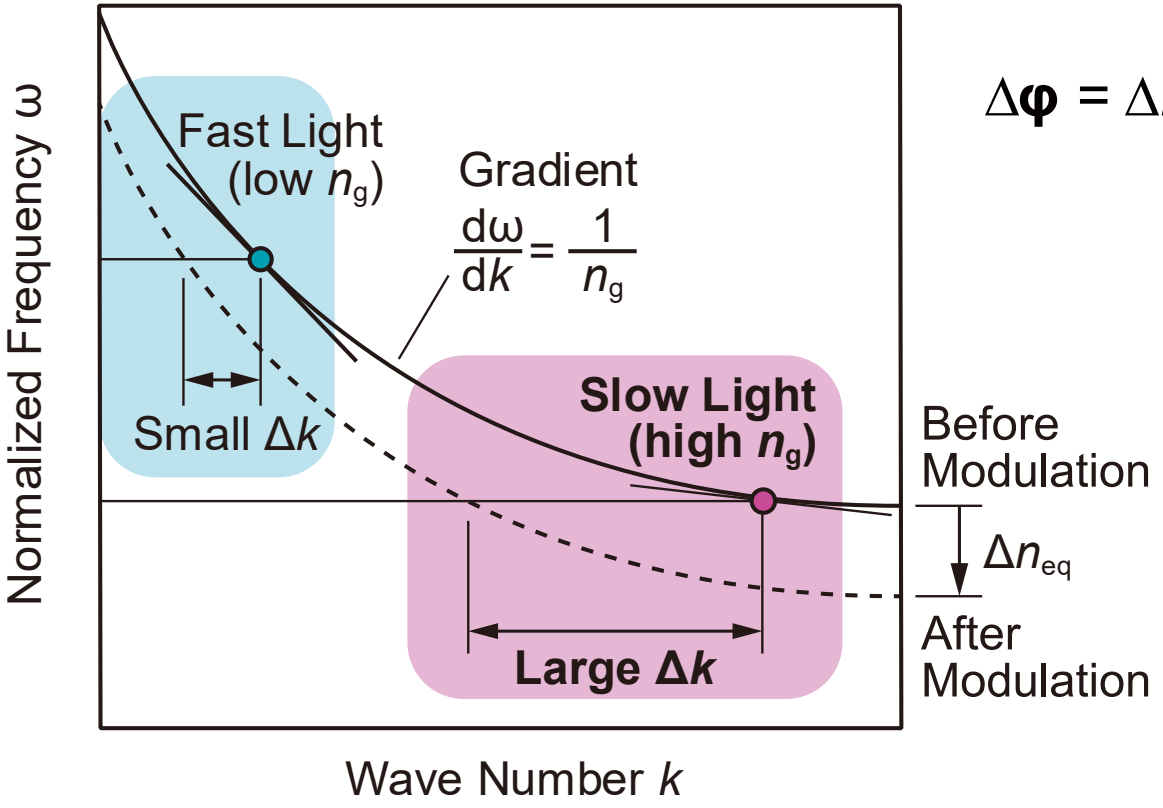
Too large n_1



ER = 10.4 dB

Slow Light Effect

Modulation efficiency of a PCW phase shifter ∝ group index n_g



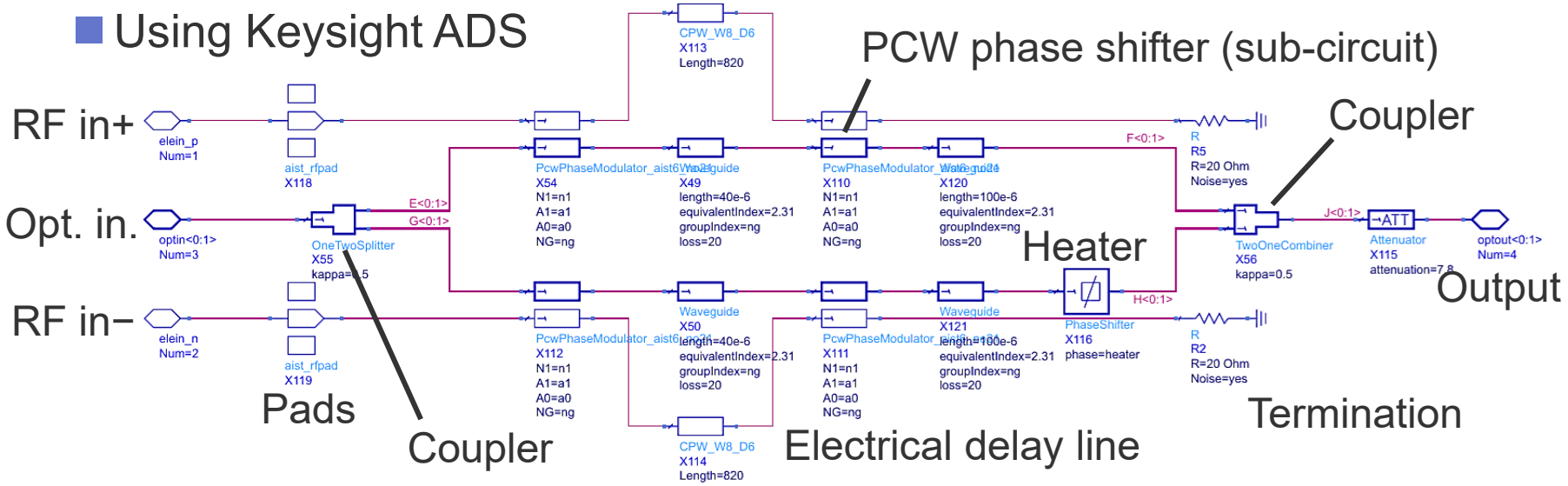
$$\Delta\varphi = \Delta kL = k_0 \boldsymbol{n_g} \zeta \frac{\Delta n_{eq}}{n_{eq}} L$$

$$\left[\zeta \equiv \frac{n_{eq}}{\omega_b} \frac{d\omega_b}{dn_{eq}} \right]$$

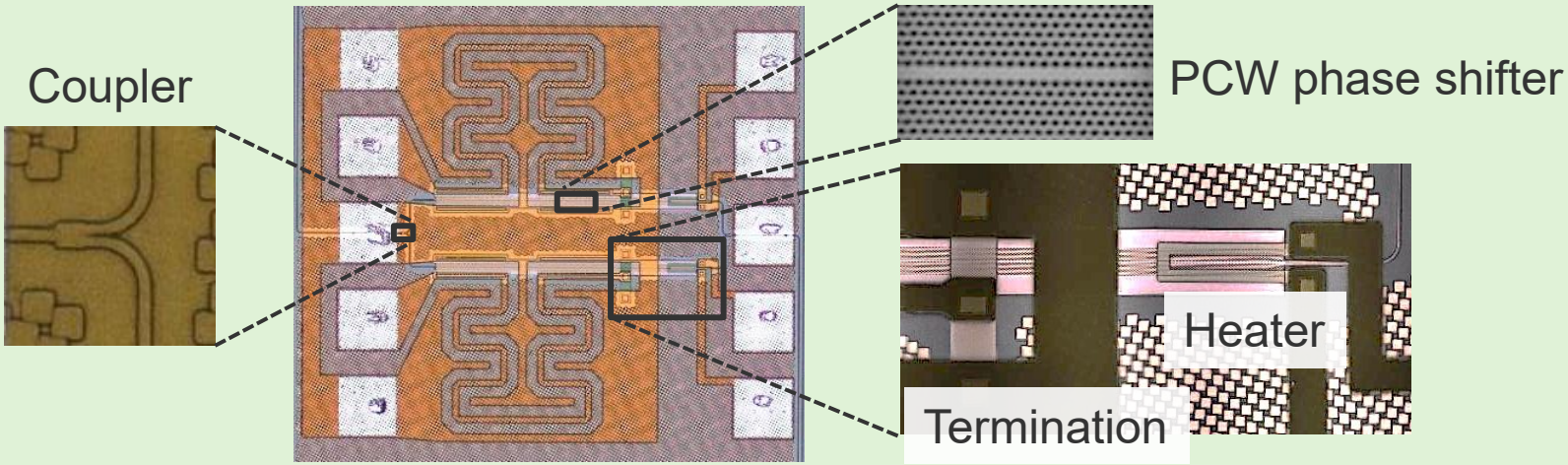
Verilog-A model equation of a PCW phase shifter: $n_{eq}(V) = n_0 + \boldsymbol{n_g} n_1 V$

Full Simulation Schematic

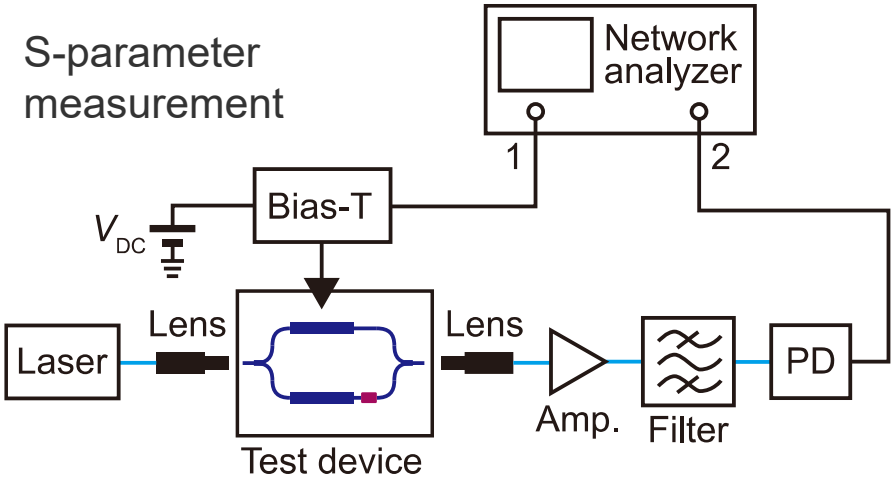
- Using Keysight ADS

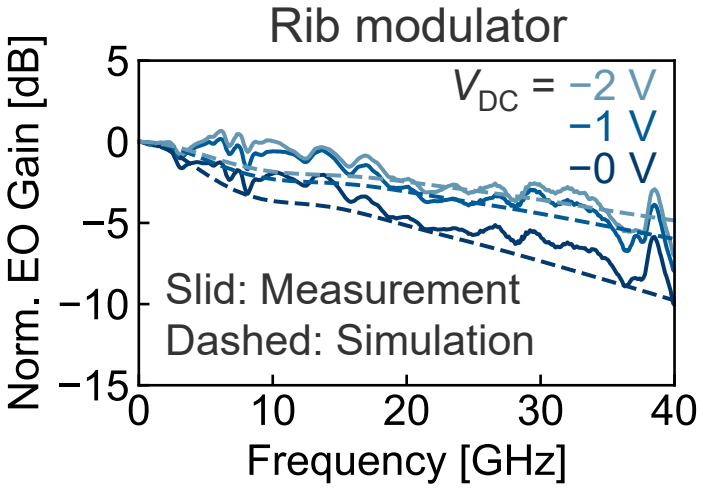


Simulation schematic corresponds perfectly to the actual device

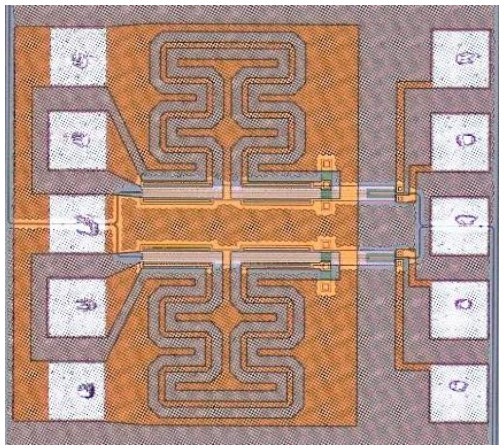


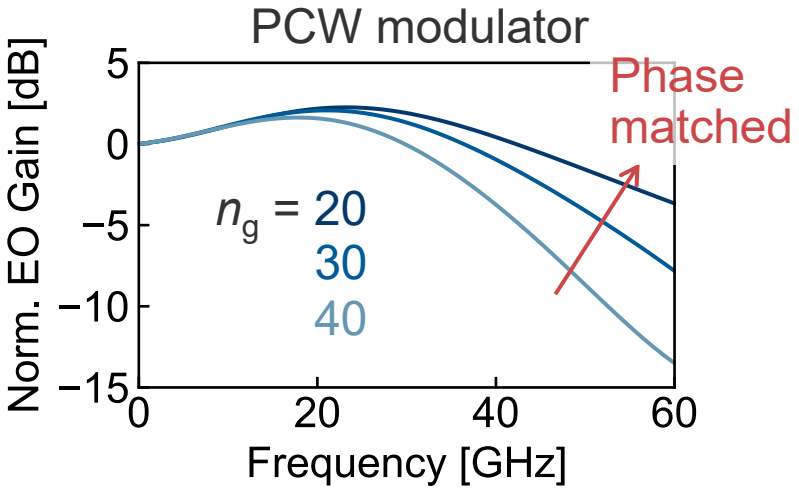
Result - Frequency Response





Simulation agreed well with the measurement up to 40 GHz



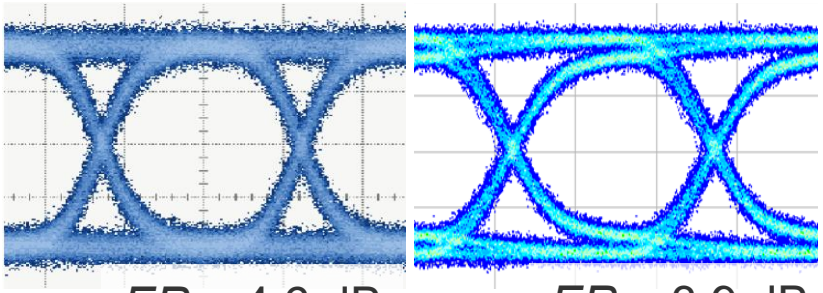


For PCW modulators, simulation can estimate n_g dependency

Eye Diagram (Rib Modulator)

Measurement Simulation

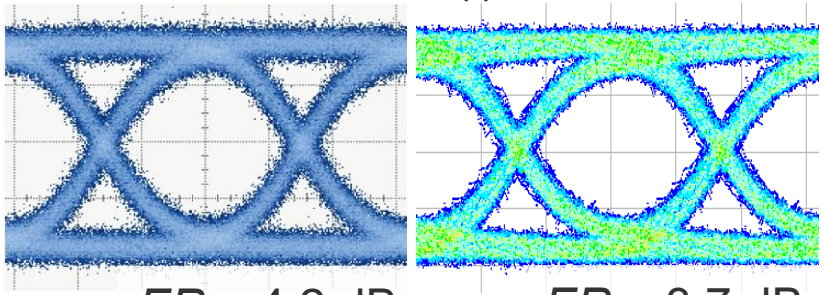
20 Gbps, V_{pp} = 2 V



ER = 4.6 dB *ER* = 3.9 dB

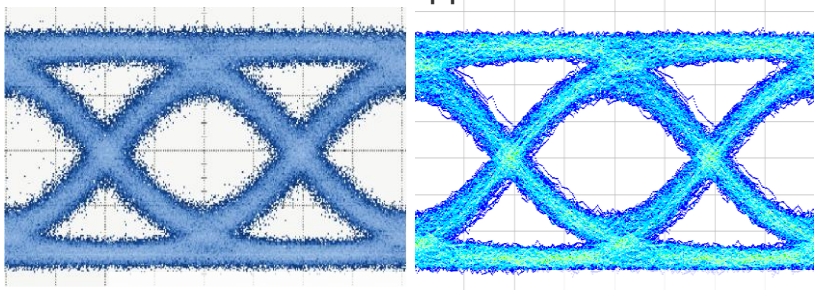
Measurement Simulation

32 Gbps, V_{pp} = 2 V



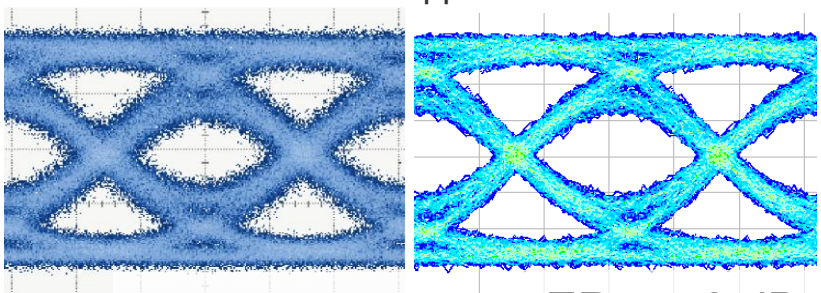
ER = 4.2 dB *ER* = 3.7 dB

50 Gbps, V_{pp} = 3.5 V



ER = 5.9 dB *ER* = 5.9 dB

64 Gbps, V_{pp} = 3.5 V

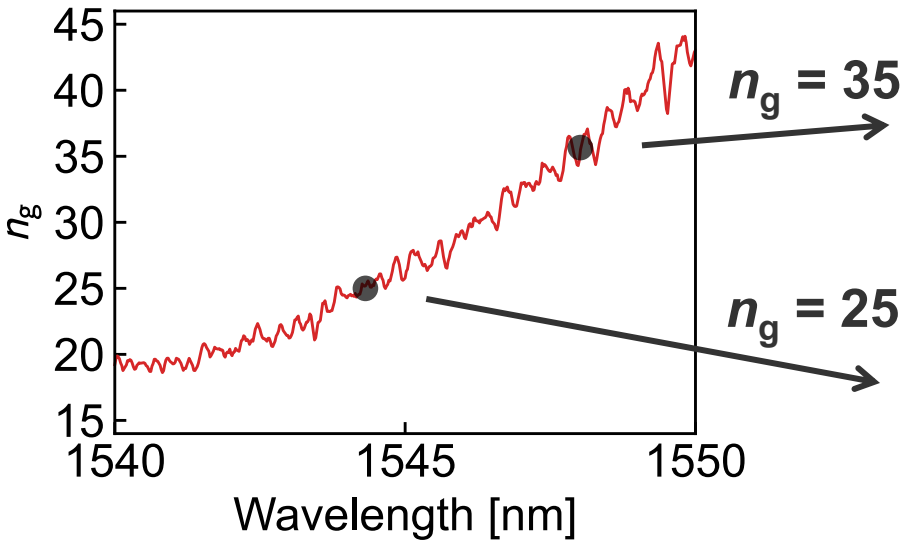


ER = 5.0 dB *ER* = 5.2 dB

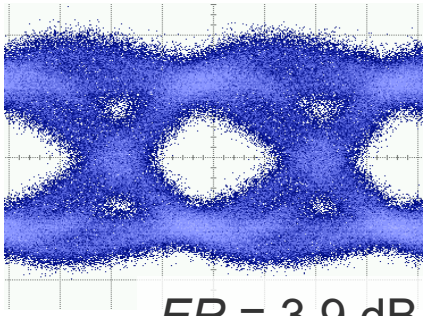
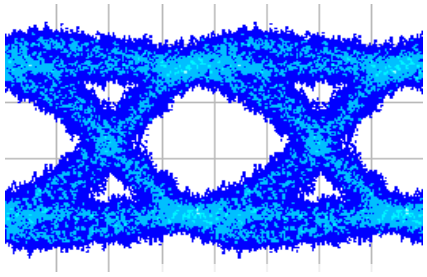
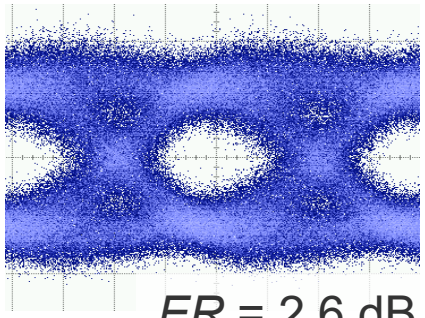
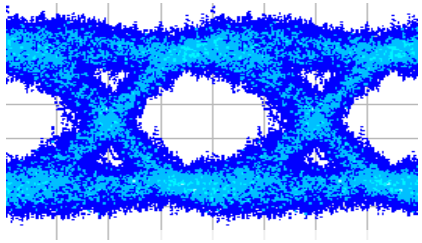
The error of *ER* was less than 1 dB up to 64 Gbps

Eye Diagram (PCW Modulator)

n_g dependence of *ER* was measured at different wavelengths



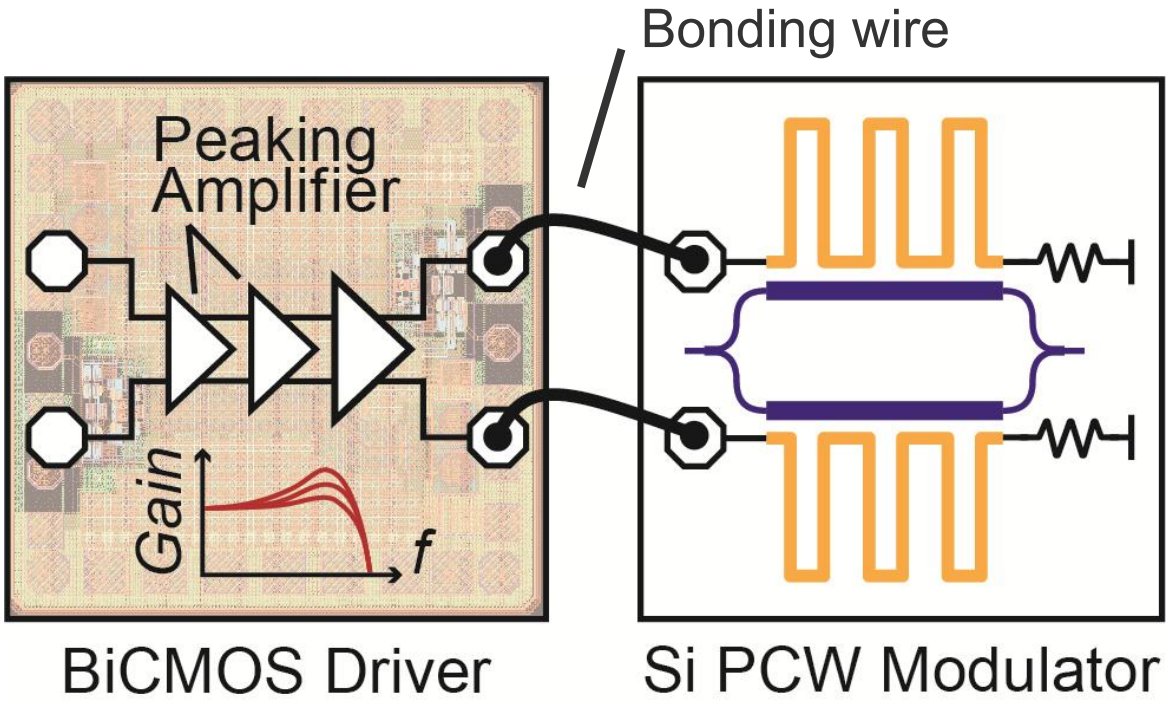
50 Gbps, V_{pp} = 3.5 V

Measurement	Simulation
	
ER = 3.9 dB	*ER* = 3.7 dB
	
ER = 2.6 dB	*ER* = 2.7 dB

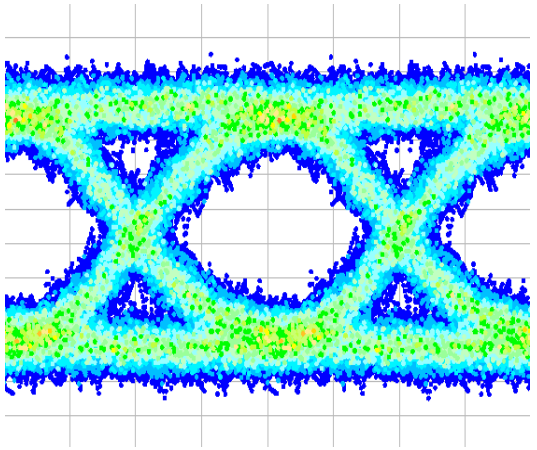
Model accurately predicted *ER* improvements due to slow light effect

EO Co-Simulation

- Hybrid integration of a SiGe BiCMOS driver and Si PCW modulator
- BiCMOS device models were provided from a foundry



Optical Out., **50 Gbps**



ER = 3.5 dB

Co-simulation predicted the overall performance

Conclusion

Purpose

- Remove the barrier between electronics and photonics
- Verilog-A modeling of Si optical modulators at high frequencies

Result

- Accurate model including propagation loss and slow light effect
- Good agreement between measurement and simulation up to 64 Gbaud

Conclusion

Verilog-A modeling approach is effective for the state-of-the-art EO co-design beyond 50 Gbaud

Future work: Process variation modeling for yield analysis